

The Art of Prompt Engineering

1. Introduction & Context

1.1. Overview

Objective:

- This lecture introduces prompt engineering—a skill essential for optimizing AI model interactions. Attendees learn how to craft, refine, and optimize prompts to achieve specific, high-quality outputs from AI systems, which is highlighted as one of the highest-paying roles in AI today.



2. Main Content

2.1. Introduction to Prompt Engineering

2.1.1. Significance of Prompt Engineering

- **Key Points:** Described as an “amazing and mind-blowing skill.” Recognized by the World Economic Forum as the highest paying AI job. The goal is to optimize AI interactions for maximum efficiency and accuracy.

2.2. Understanding Prompt Engineering

2.2.1. What Are Prompts?

- **Definition:** Prompts are the instructions, context, or input provided to AI models to perform specific tasks.
- **Purpose:** They guide AI models to generate relevant and accurate responses.

2.2.2. Prompt Engineering Defined

- **Definition:** The art of crafting and optimizing prompts to achieve the best possible outputs from AI.
- **Key Challenge:** Addressing issues where AI misinterprets vague instructions and delivers unexpected results.
- **Objective:** To learn how to write detailed, specific, and context-rich prompts.



2.3. Agenda for the Session

- Introduction to Prompt Engineering
- Why Prompt Engineering Matters for Professionals
- Key Concepts in Prompt Engineering
- Designing Prompts for Real-World Tasks
- Advanced Techniques for Better Results
- Practical Tips and Strategies for Generating Good Prompts
- Tools and Templates for Generating Prompts
- **Outcome:** By the end of the session, attendees will be able to create optimized prompts for any available Large Language Model (LLM).

2.4. Goal for Today

- **Primary Goals:** Learn structured, optimized, and advanced prompting techniques. Understand that well-structured prompts improve AI accuracy and relevance.

2.5. Examples: Bad Prompt vs. Good Prompt

Bad Prompt:

"Explain marketing strategy"

Issue: Too vague, lacks specifics.

Good Prompt:

"You are a marketing consultant; provide 5 marketing strategies for small businesses, emphasizing low-cost digital marketing methods."

Benefit: Specific context and clear instructions lead to better, targeted outputs.

Bad AI Prompts

Turn person into a cat



Good AI Prompts

Turn subject in photo into a cute tabby cat with brown and black fur in a watercolor style painting, include starry night sky behind subject

Make image space themed



Create hyper-realistic space themed image, person is wearing an astronaut uniform, colorful galaxies and shooting stars in the dark moody sky

A cool colorful painting



A super colorful and bold painting style similar to pop art styles by Andy Warhol with detailed paint brush strokes and geometric shapes in the background of the subject

2.6. Types of Prompt Engineering

2.6.1. Context and Constraint-Based Prompting

- **Definition:** Using specific background information (context) and clear boundaries (constraints) to guide AI.
- **Importance:** Reduces ambiguity and tailors outputs to user needs.
- **Examples:**
 - *Product Description:* Weak Prompt: "Write a product description for a smartphone." Improved Prompt: "Write a product description for the XYZ pro 12 smartphone highlighting its 108-megapixel camera, 120 hertz OLED display, and 5,000 mAh battery. The tone should be engaging and appealing to tech enthusiasts."
 - *Translation:* Weak Prompt: "Translate the sentence." Strong Prompt: "Translate the following sentence from English to Hindi, ensuring the cultural nuances are preserved: 'The Festival of Lights brings joys to everyone.'"
 - *News Article:* Weak Prompt: "Write a news article." Strong Prompt: "Write a 300-word news article suitable for the Times of India about the recent advancements in AI technology in Bangalore, focusing on local startups."
- **Real-World Applications:** Customer support emails, educational content, and healthcare communication.

2.6.2. Role-Based Prompting

- **Definition:** Assigning the AI a specific role to tailor its responses.
- **How It Works:** Simulates expertise, adjusts tone and style, and enhances engagement.
- **Examples:**
 - *Interview Questions:* Without Role-Based Prompting: "Give me answers for a few common interview questions." With Role-Based Prompting: "As an experienced HR professional, craft well-structured answers for common interview questions. Make them confident, concise, and aligned with best hiring practices."
 - *Resume Writing:* Weak: "Help me write a resume." Strong: "As a professional career coach, craft a compelling resume for a software engineer with 5 years of experience in Python and AI, ensuring it aligns with current hiring trends."
 - *Investment Advice:* Combining role-based and context/constraint-based prompting for tailored advice.

2.6.3. Zero-Shot and Few-Shot Prompting

- **Definition:** Zero-Shot: Instructing AI without examples. Few-Shot: Providing a few examples before asking the task.
- **Why It Works:** Few-shot prompting helps AI recognize patterns, ensuring responses align with expectations.
- **Examples:**
 - *Product Descriptions:* Zero-Shot: "Write a product description for a smartphone." Few-Shot: Provide two examples of premium smartphone descriptions, then ask for a tailored description.
 - *Social Media Posts:* Provide examples of posts before asking AI to generate a new one in a similar style.

2.6.4. Advanced Prompting Techniques

2.6.4.1. Chain of Thought (CoT) Prompting

- **Definition:** Instructing AI to explain its reasoning step by step before providing the final answer.

- **Examples:**

- *Customer Complaint Response:* General: "Write a response to a customer complaining about a delayed order." CoT: "Generate a professional and empathetic response to a customer frustrated about a delayed order. Structure it step by step: 1. Acknowledge the customer's frustration; 2. Apologize; 3. Explain the cause of the delay; 4. Suggest next steps."
- *Investment Advice Analysis:* "Analyze if a middle-class investor in India should allocate 10 lakhs in mutual funds. Break it down step by step."

2.6.4.2. Tree of Thought (ToT) Prompting

- **Definition:** Encourages AI to explore multiple reasoning pathways before arriving at a final answer.
- **Example:** "Analyze the pros and cons of three possible market entry strategies for international expansion: Joint Venture, Direct Investment, and Franchising. Recommend the best strategy based on scalability and risk."
- **Benefit:** Provides comprehensive analysis and aids in decision-making where multiple solutions exist.

2.6.4.3. Self-Consistency Prompting

- **Definition:** Requesting multiple independent responses to the same prompt and selecting the most consistent answer.
- **Examples:** Generating several sets of employer retention strategies or salary negotiation tactics, then summarizing common effective strategies.
- **Benefit:** Reduces errors by considering multiple viewpoints.

2.6.4.4. ReAct Prompting (Reason + Act + Reflection)

- **Definition:** Involves prompting AI to reason through a problem, act by providing a step-by-step answer, and reflect to ensure completeness.
- **Examples:** Business Report Summarisation—"Analyze this report using the following steps: 1. Identify key trends; 2. Extract funding patterns; 3. Verify all critical insights are covered; 4. Summarize the findings."
- **Benefit:** Ensures logical and structured responses with fewer omissions.

2.6.4.5. Multimodal Prompting

- **Definition:** Combining multiple input formats (text, images, audio, etc.) in one prompt.

- **Examples:** "Upload an image of the Taj Mahal and prompt: 'Analyze this image and provide a historical summary.'"
- **Real-World Applications:** Marketing (analyzing ad images), education (explaining diagrams), UI/UX design, and data analysis.

ADVANCED PROMPT ENGINEERING TECHNIQUES

```
Article: {{ ARTICLE }}
```

You will generate increasingly concise, entity-dense summaries of the a
Repeat the following 2 steps 5 times.

Step 1. Identify 1-3 informative Entities (" "; " delimited) from the Art
Step 2. Write a new, denser summary of identical length which covers ev

A Missing Entity is:

- Relevant: to the main story.
- Specific: descriptive yet concise (5 words or fewer).
- Novel: not in the previous summary.
- Faithful: present in the Article.
- Anywhere: located anywhere in the Article.

Guidelines:

- The first summary should be long (4-5 sentences, ~80 words) yet highl
- Make every word count: rewrite the previous summary to improve flow a
- Make space with fusion, compression, and removal of uninformative phr
- The summaries should become highly dense and concise yet self-contain
- Missing entities can appear anywhere in the new summary.
- Never drop entities from the previous summary. If space cannot be mad

Remember, use the exact same number of words for each summary.

Answer in JSON. The JSON should be a list (length 5) of dictionaries wh

2.7. Real-World Applications and Use Cases

2.7.1. Industry-Specific Examples

- **Product Requirement Analysis (Software Development):** Role-based prompting with Chain of Thought to break down features.
- **Competitive Market Analysis (Marketing & Strategy):** Self-Consistency prompting for generating multiple analyses.
- **Customer Support:** Role-based plus context & constraint-based prompting to draft precise responses.
- **Investment Portfolio Strategy (Finance):** Using Chain of Thought prompting for detailed step-by-step analysis.
- **Product Launch Content (Marketing):** Few-Shot prompting combined with role-based prompting.
- **Bug Troubleshooting (IT Support):** ReAct prompting for structured problem solving.

2.7.2. Resources for Learners

- **Prompt Engineering Handbook:** A summary of session content, best practices, and prompting techniques ([Access Here](#)).
- **Ready-to-Use Prompt Formulas:** Pre-made templates for various tasks (content writing, HR, emails, etc.) ([Download Templates](#)).
- **Prompt Generator (Jarvis):** An interactive tool to generate advanced-level prompts through clarifying questions and tailored outputs ([Try It Now](#)).

3. Key Insights & Takeaways

- **Fundamental Understanding:** Prompt Engineering is an essential skill in optimizing AI interactions for quality and efficiency.
- **Types & Techniques:** Different prompting strategies (context-based, role-based, zero/few-shot, CoT, ToT, Self-Consistency, ReAct, and multimodal) cater to various needs and outputs.
- **Practical Impact:** Well-defined and optimized prompts can transform generic AI responses into highly tailored outputs.
- **Real-World Applications:** The techniques apply across industries—from customer support to investment advice—demonstrating versatility.
- **Resources:** Learners gain exclusive access to handbooks, prompt templates, and prompt generators that further enhance their prompt engineering skills.

4. Conclusion & Summary

4.1. Summary of Core Message

Core Message: The session provided a comprehensive overview of prompt engineering techniques. By exploring various methods—from context-based and role-based prompting to advanced strategies like Chain of Thought and multimodal prompting—attendees learn how to refine AI interactions for better accuracy, relevance, and efficiency.

4.2. Next Steps & Further Reading

- **Next Steps:** Practice implementing these prompting techniques with real-world tasks. Experiment with the provided resources (handbook, prompt formulas, prompt generator) to master prompt engineering.
- **Further Reading:** Additional materials on AI prompting strategies. Relevant articles and case studies on advanced prompt engineering in professional settings.

5. Additional Enhancements for Readability & Engagement

5.1. Comparison Table: Prompting Techniques

Technique	Definition	Key Benefit
Context & Constraint-Based	Provides detailed background and sets clear boundaries	Reduces ambiguity and tailors output to specific needs
Role-Based Prompting	Assigns a specific role to the AI	Adjusts tone, expertise, and style for more professional

Technique	Definition	Key Benefit
		output
Zero-Shot/Few-Shot	Instructs with no or few examples to guide response format	Helps AI recognize patterns and align output with expectations
Chain of Thought (CoT)	Requires step-by-step reasoning before the final answer	Enhances logical reasoning and clarity in complex tasks
Tree of Thought (ToT)	Explores multiple reasoning paths before finalizing an answer	Useful for decision-making in complex scenarios
Self-Consistency	Generates multiple responses to select the most consistent answer	Reduces errors by ensuring a consensus across responses
ReAct Prompting	Involves reasoning, acting, and reflecting in a structured process	Ensures comprehensive and well-verified output
Multimodal Prompting	Uses multiple input types (text, image, etc.) for richer responses	Enhances understanding through diverse data formats

5.2. Self-Check Q&A

- Q: What is prompt engineering and why is it important?
- Q: How does role-based prompting improve AI responses?
- Q: What is the key difference between zero-shot and few-shot prompting?